Sean Balbale

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SUMMARY

ECE/CS student at Trinity College building across the stack — bare-metal firmware, enterprise Java backends, and a self-hosted AI product running on infrastructure I designed and operate myself. I care about precision under real constraints, whether that's a sub-100ms embedded control loop or a production Spring Boot migration with zero downtime.

EDUCATION

Trinity College — Hartford, CT *Expected May 2027*
Bachelor of Science in Electrical and Computer Engineering GPA: 3.70
Bachelor of Science in Computer Science
Coursework: High-Performance Computing, Embedded Systems Design, Deep Learning, Software Engineering, Microelectronic Circuits, Data Structures & Algorithms, Automatic Control Systems, Intro to Computer Systems.
Extracurriculars: Varsity Athlete — Trinity College Men's Rowing; Singer & Technology Chair — The Trinity College Accidentals.

SKILLS

Languages: Java, Python, C, C++, TypeScript, JavaScript, Assembly (8051/MIPS), Verilog, SQL, MATLAB, LaTeX
Backend: Spring Boot 3, Jakarta EE, JUnit 5, Mockito, REST APIs, Node.js, FastAPI
Frontend: React, Next.js, Svelte, Tailwind CSS, AngularJS
Cloud & DevOps: AWS, Kubernetes, Docker, Proxmox, Git, Linux, CI/CD, Supabase, Slurm, MPI
Embedded / Hardware: Arduino, ARM, 8051, FPGA, BLE 5, ANT+, SPI/I2C, LTspice, Oscilloscopes, Logic Analyzers
Other: SolidWorks, 3D Printing (Klipper), Precision Metrology, Systems Engineering

WORK EXPERIENCE

Cloud Service Software Engineering Intern — Sonos | Boston, MA *Summer 2026*
- Led migration of two production Java/Spring Boot services from WebSocket command delivery to an SQS-backed queue: designed the feature-flagged per-player fan-out with retry-until-acceptance semantics, and validated both transports against real speakers.
- Diagnosed a silent delivery failure in pre-production by tracing commands through the SDK and queue infrastructure to an endpoint-routing defect, and drove the fix across both services.
- Made the functional-test suite runnable locally by porting its DynamoDB-stream Lambda into a local harness for pre-push validation; also built Datadog dashboards and monitors for the new delivery path.

Engineering Teaching Assistant — ENGR 212L Linear Circuit Theory, Trinity College | Hartford, CT *Spring 2026*
- Led weekly lab sessions for ENGR 212L covering RLC circuit analysis, op-amps, Laplace transforms, Fourier series, and active filter design.

Software Engineering Intern — Bullhorn | Boston, MA *Summer 2025*
- Owned BH Boot 1.0.0 framework migration: upgraded multiple production Java services from Spring Boot 2 to Spring Boot 3, refactoring all `javax` namespaces to `jakarta` and restructuring the proprietary data access layer.
- Led Log4j2 security upgrade across SEARCH_AI, ATS_CRM, and MIDDLE_OFFICE groups; refactored legacy TestNG/JMockit test suites to JUnit 5 and Mockito; managed feature branches, merge requests, and peer code reviews.

Metrology Intern — East Coast Metrology, A Division of In-Place Machining Co. | Topsfield, MA *Summer 2024*
- Calibrated portable CMMs, total stations, laser trackers, and large-volume scanners; verified manufactured parts against SolidWorks CAD tolerances in-house and on-site.

RESEARCH, HONORS & SELECTED PROJECTS — MORE ON PORTFOLIO WEBSITE

"Embodied Harms and Inferred Data: Redefining Privacy in Extended Reality" *Expected Early 2027*
The International Journal of Law, Ethics, and Technology — In Production
- Peer-reviewed paper arguing BIPA and GDPR are structurally inadequate for XR; proposes a "Bodyright" framework treating biometric data as an inalienable property right, not a consent-revocable one. Awarded the **Alumni Prize in English Composition**, Trinity College, May 2026.

Astraphe AI — Mobile AI Athletic Coaching Platform *2026*
- Built a mobile-first AI coaching platform: multi-source data pipeline (Strava, WHOOP, Garmin, intervals.icu) deduplicated onto canonical records; a server-side training-load engine (TSS, CTL/ATL/TSB, Strain) in FastAPI; a Gemma 4 coaching agent with a pgvector RAG pipeline over coaching memory.
- Svelte 5 / Capacitor mobile client (iOS/Android), self-hosted on Kubernetes (k3s) with CI/CD via GitHub Actions (80% coverage gate); added an MCP server exposing user data to Claude via OAuth 2.1, open-sourced under Apache-2.0.

Cherrybrook Homelab — Self-Hosted Infrastructure *2025 – Present*
- Built and operate a four-node Proxmox VE cluster (i7-9700K + Tesla V100 server, two Dell PowerEdge servers, a ThinkPad quorum/backup node) hosting Dockerized production workloads including Astraphe AI and a self-hosted Second Brain MCP server.
- Executed a live VLAN/subnet migration with split-horizon DNS to resolve VPN conflicts; manages reverse proxy/SSL, tailnet DNS, and a UniFi network stack.

PERSONAL

Serious home cook across a wide range of cuisines — a proper katsu ramen takes hours and a good samosa crimp takes thirty seconds of muscle memory. Outside the kitchen: cycling, ski mountaineering, and tinkering on whatever project has captured my attention that week.